Department of Information Systems
Bachelor of Information Systems
Lab 1
IS301 Structured Programming

Weighting: 4%

ULO addressed: i, ii

Due date: Friday 28th February 2025 @11:59PM

Instructions:

1. Download template.c file (attached together with this document) and rename it to the following format: *first initial of your first name, followed by your last name, an underscore character followed by lab1*. Example: rpur_lab1.c.
2. Develop a/an pseudocode/algorithm to solve this problem within the code.
3. Write your C code to solve the above problem.
4. Compile and run your code. Your code should be without errors. Take Screen-shots/ snipped images of the results from the different data sets provided.
5. Once done, submit a single zipped file containing the C code, and images (consider having this in a folder). The zipped file name should have same format as above.
6. submit the zipped file before due date.

Problem:

Write a program to create a customer's bill for a company. The company sells only five different products. TV, Monitor, Flash Drive, Hard Drives, DeskJet Printer. The unit price is K400, K220, K35.20, K150 and K300, respectively. The program must read the quantity of each piece of equipment purchased from the keyboard. It then calculates the cost of each item, the subtotal and the total cost after an 8.35% sales tax.

The input data consist of a set of integer representing the quantities of each item sold. These integers much be input into the program in a user-friendly way; that is, the program must prompt the user for each quantity as shown below. The numbers in the boldface shows the user's answers.

How Many TVs Were Sole? **3**
How Many Monitors Were Sold? **5**
How many Flash Drives Were Sold? **1**
How many Hard Drives Were Sold? **2**
How many Deskjet Printers Were Sold? **4**

The output should be shown as following:

```
QTY     DESCRIPTION    UNIT PRICE      TOTAL PRICE
-------------------------------------------------------------------
xx      TV             400.00          xxxx.xx
xx      Monitor        220.00          xxxx.xx
xx      Flash Drive     35.20          xxxx.xx
xx      Hard Drive     150.00          xxxx.xx
xx      Deskjet Printer 300.00         xxxx.xx
                                     ----------------
                       Subtotal:      xxxxx.xx
                       Tax:            xxxx.xx
                       Total:         xxxxx.xx
-------------------------------------------------------------------
```

Notes:

- Use either defined constants or memory constants for the unit prices and the tax rate.
- Use integer variables to store the quantities for each item.
- Use floating-point variable to store the total price, the bill subtotal, the tax amount, and the total amount of bill.
- Run you program twice with the following data:

SET1	2 1 4 1 2
SET2	3 0 2 0 21